Q.22 What are the advantages and disadvantages of wet cleaning?

Q.23 How we can calculate the crushing efficiency

Q.24 How moisture and hardness affect the efficiency of size reduction

Q.25 Explain the working of color sorter with diagram

Q.26 Describe advantages of size reduction

Q.27 What are the applications of screening

Q.28 Which factors affects the effectiveness of mixing

Q.29 Write a note on sedimentation

Q.30 Write a note on filter aid filtration

Q.31 Define the grinding process? Draw a diagram of hammer mill showing different parts

Q.32 Write a short note on effect of size reduction on sensory characteristics and nutritive value of food?

Q.33 What are the different types of screen?

Q.34 Write a note on magnetic cleaning?

Q.35 Write a brief note on crystallization?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Name different types of mixing equipment and explain at least two with diagram?

Q.37 Define the distillation process? Explain any two types of distillation process?

Q.38 Draw a neat and clean diagram of sedimentation tank and discuss different parts?

Roll No. ....

3rd Sem / Food Technology

Sub : Unit Operations in Food Processing/Pr. of Food Engg.

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Which of these is a 'dry' cleaning method?

a) Soaking

b) Magnetic separation

c) Spray washing

d) Ultrasonic cleaning

Q.2 Separating the fruits by color using electronic sensors is an example of:

a) Sorting b) Cleaning

c) Agitation d) Distillation

Q.3 Preliminary unit operations are usually performed:

a) After packaging

b) During distribution

c) At the start of the process

d) Only for liquid foods

Q.4 Agitation refers to:

a) Creating a uniform mixture of powder

b) Inducing motion in a materials in a specified way

c) Separating solids from liquids

d) Reducing particle size

Q.5 Mixing of two miscible liquids is often called:

a) Emulsification b) Blending

c) Dispersion d) Kneading

Q.6 Which force is dominant in a Jaw Crusher?

a) Impact b) Shearing

c) Compression d) Attrition

Q.7 A "Centrifuge" accelerates which process?

a) Distillation b) Sedimentation

c) Size reduction d) Mixing

Q.8 The "Filter Cake" refers to:

a) The equipment used for filtering

b) The accumulated solids on the filter medium

c) The final liquid product

d) A type of dessert

Q.9 "Flocculation" is often used to aid which process?

a) Sedimentation / Clarification

b) Distillation

c) Sieve analysis

d) Milling

Q.10 Sieving efficiency decreases with:

a) Dry feed b) Low feed rate

c) Vibration d) High Moisture

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 For fibrous foods, size reduction is usually achieved by _____.

Q.12 In sieving, the number of openings per linear inch is known as the ____.

Q.13 A Sigma Blade mixer is specifically designed for kneading high-viscosity doughs. (True / False)

Q.14 The liquid that passes through a filter medium is called the _____.

Q.15 Distillation separates components based on differences in their Volatility or boiling points. (True / False)

Q.16 The formation of solid particles from a homogeneous solution is called_____.

Q.17 Sedimentation does not use the force of Gravity to separate solids from liquids. (True / False)

Q.18 _____ Mills use both impact and Attrition to grind materials into fine powder.

Q.19 Screening efficiency is 100% in industrial setups. (True / False)

Q.20 _____ is used to separate cream from milk.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What are the aims and applications of grading?